

FinPilot - Developer Onboarding & Learning Guide

Welcome to the FinPilot (AI Tax Assistant) project! This document will guide you step-by-step on how to learn this project, understand its architecture, and the exact order in which you should explore the files.

Phase 1: Project Overview & Architecture

The project is divided into two main parts:

1. Backend: A Python Django REST Framework application providing APIs.
2. Frontend: A React application built with Vite and TailwindCSS.

Start by looking at the root folders: 'backend/' and 'frontend/'.

Check 'backend/requirements.txt' to see the Python dependencies, and 'frontend/package.json' for the JS dependencies.

Phase 2: Deep Dive into the Backend (Django)

The backend is modularized into different 'apps'. Here is the order to explore them:

1. backend/config/ (or core app):
 - Look at 'settings.py' to understand database configurations (PostgreSQL, MongoDB) and installed apps.
 - Look at 'urls.py' to see the main API routing.
2. backend/apps/accounts/ & companies/:
 - Check 'models.py' to understand how Users and Companies are structured. Everything ties back to a Company.
3. backend/apps/transactions/ & documents/:
 - This is where the core data is handled. Understand how income, expenses, and uploaded invoices are stored.
4. backend/apps/tax/ (Crucial Business Logic):
 - Start with 'rules.py', 'gst_rules.py', and 'tds_rules.py'. These contain pure python logic for tax formulas.
 - Then look at 'services.py' to see how these formulas are applied to the database records.
5. backend/apps/ml/ & analytics/:
 - Explore 'services.py' here to see how AI features (like document parsing or categorizing transactions) and dashboard analytics are implemented.

File exploration pattern for any Django app:

- > First look at 'models.py' (Database schema)
- > Then 'serializers.py' (Data formatting)
- > Then 'views.py' & 'urls.py' (API Endpoints)
- > Finally 'services.py' or 'utils.py' (Business Logic)

Phase 3: Deep Dive into the Frontend (React)

The frontend consumes the APIs you just explored. Here is how to navigate it:

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1. Configuration:

- Check 'vite.config.js' and 'tailwind.config.js' for build and styling setups.

2. Entry Points:

- Look at 'src/main.jsx' and 'src/App.jsx'. See how the app is wrapped in providers (Auth, Theme) and how the Router is set up.

3. src/api/ or src/services/:

- Check how API calls are made (usually via Axios or Fetch). See how it connects to the Django backend URLs.

4. src/pages/ or src/views/:

- Explore the main screens (e.g., Dashboard, TaxEstimator, Transactions list).
- See how they fetch data from the API and pass it down.

5. src/components/:

- Explore reusable UI components (Buttons, Tables, Modals, Charts). These are mostly styled using TailwindCSS.

6. src/context/ or src/store/:

- Understand how global state like User Authentication or current selected Company is managed across the app.

Summary: How to trace a feature

To truly understand the flow, try tracing a single feature (e.g., 'Estimating Tax'):

1. Find the React component in 'frontend/src/pages' where the user clicks 'Estimate'.
2. Trace the API call in 'frontend/src/api' to see what URL is hit.
3. Find that URL in 'backend/apps/tax/urls.py'.
4. Look at the mapped function in 'backend/apps/tax/views.py'.
5. Follow the logic down to 'services.py' and 'rules.py' to see the exact math.
6. Follow the response back up to the frontend UI.